Exp. No: 09

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Design a simple Topology and configure one Router, Two switches and PCs Using Cisco Packet Tracer

Aim:-

To design and configure a simple network topology consisting of one router, two switches, and four PCs. Using cisco Packet Tracer, and verify communication between all devices.

Procedure:-

Step 1:- Open cisco Packet Tracer

1. Click Start
2. Open cisco Packet Tracer
3. Wait for the workspace to load

Step 2:- Add Devices

- Drag 1 Router (Cisco 1941 or 2911)

Step 3:- Connect the Devices.

Use the copper straight-through cable.

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Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time=1ms TTL=127
Reply from 192.168.2.2: bytes=32 time=1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.3: bytes=32 time<1ms TTL=127
Reply from 192.168.2.3: bytes=32 time<1ms TTL=127
Reply from 192.168.2.3: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time=13ms TTL=255
Reply from 192.168.2.1: bytes=32 time=1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 13ms, Average = 3ms
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Router Configuration:-

Step 4:- open Router CLI

1. Click the Router
2. Select the CLI tab.
3. Press Enter when Prompted

step 5:- Configure Interface G0/0

Router > enable

Router #

Router (config)

Router (config -if)

Router (config - if) #

Router (config - if) # exit

Step 6:- configure Interface G0/1

interface gigabit Ethernet 0/1

ip address

no shutdown

exit

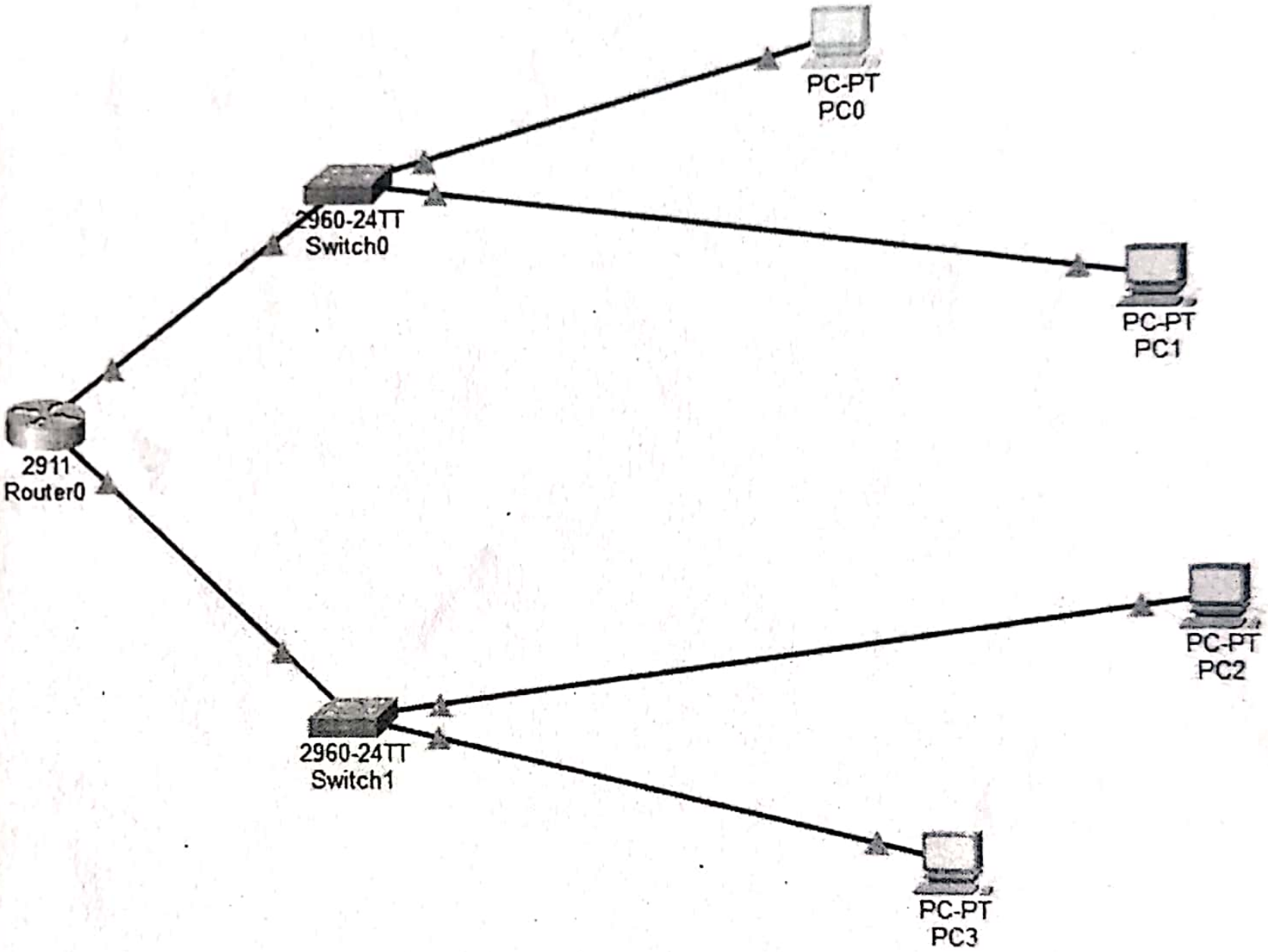
Step 7:- Save Router configuration

end

copy running

Enter when Prompted.

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PC configuration

step 8:- configure PC 0

step 9:- configure PC1

step 10:- configure PC2

step 11:- configure PC 3

step 12:- Test connectivity

Expected output:-

Packet sent = 4

Received = 4

Lost = 0

Result:- The network topology was successfully designed in cisco Packet Tracer using one router, two switches, and PCs. The router and switches were configured, and connectivity between the devices was verified successfully.

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